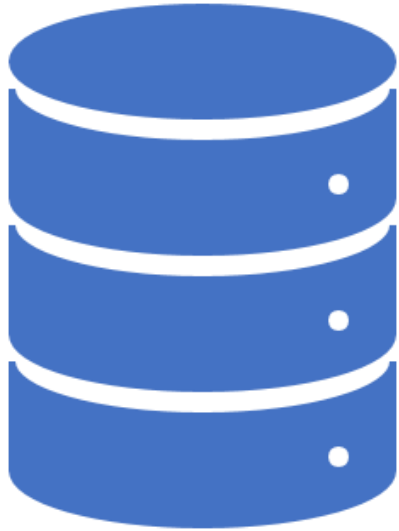




COM 736 Database Systems and Data
Analytics

Data and data storage

Objectives

Data

- What is Data?

Data Storage

- File Systems – storage and issues
- Databases – storage and advantages

Data

- Facts that can be recorded
 - Name
 - Telephone number
 - Address
- Can be recorded / stored in different ways
 - Physical Storage:
 - Paper
 - Files
 - Digital Storage:
 - Files
 - Database

Small Business / Organisations

- One person in total control of ALL business data
 - Deals with invoices/ledgers etc.
 - Has overview of everything that happens in the company
- As business / Organisation gets larger
 - A single person cannot control all the data
 - Different functions performed by different people
 - Makes management decisions more difficult

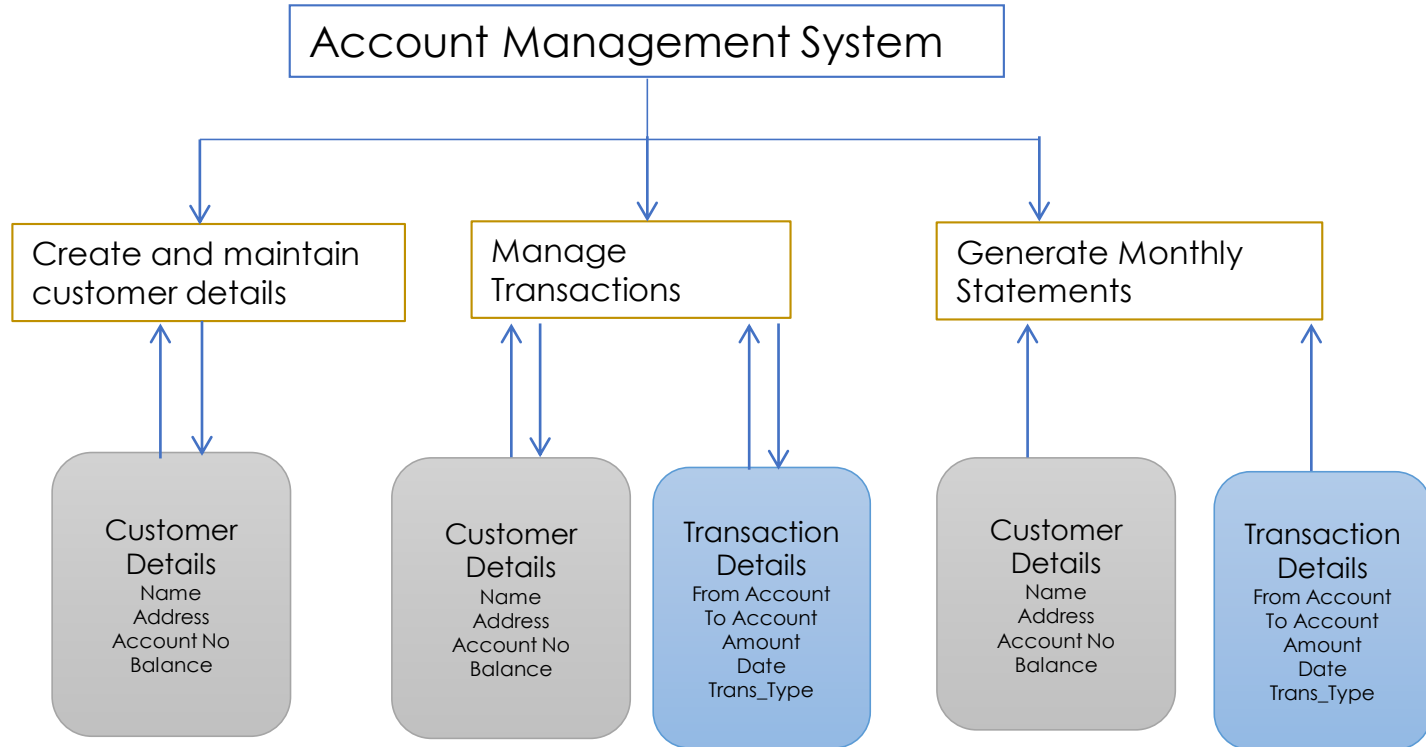
File Systems

- The basic tool for controlling disk storage
- A file is a group of bits
- Each application program places its own interpretation on the bits
- COBOL records
- Batch Processed

Problems with File Systems

- Structure of records defined by application program
 - Relationships between data items to be managed by application program
- Files from different applications often incompatible
 - Data often not transferable between applications
- Data often duplicated in different applications
- Difficult to provide generalised querying and maintenance facilities

Data duplication in a file system



Problems with file systems

- Data Duplication
- Inconsistent Data
- Data types and relationships are defined and managed through (individual) Program
- Lack of accurate information

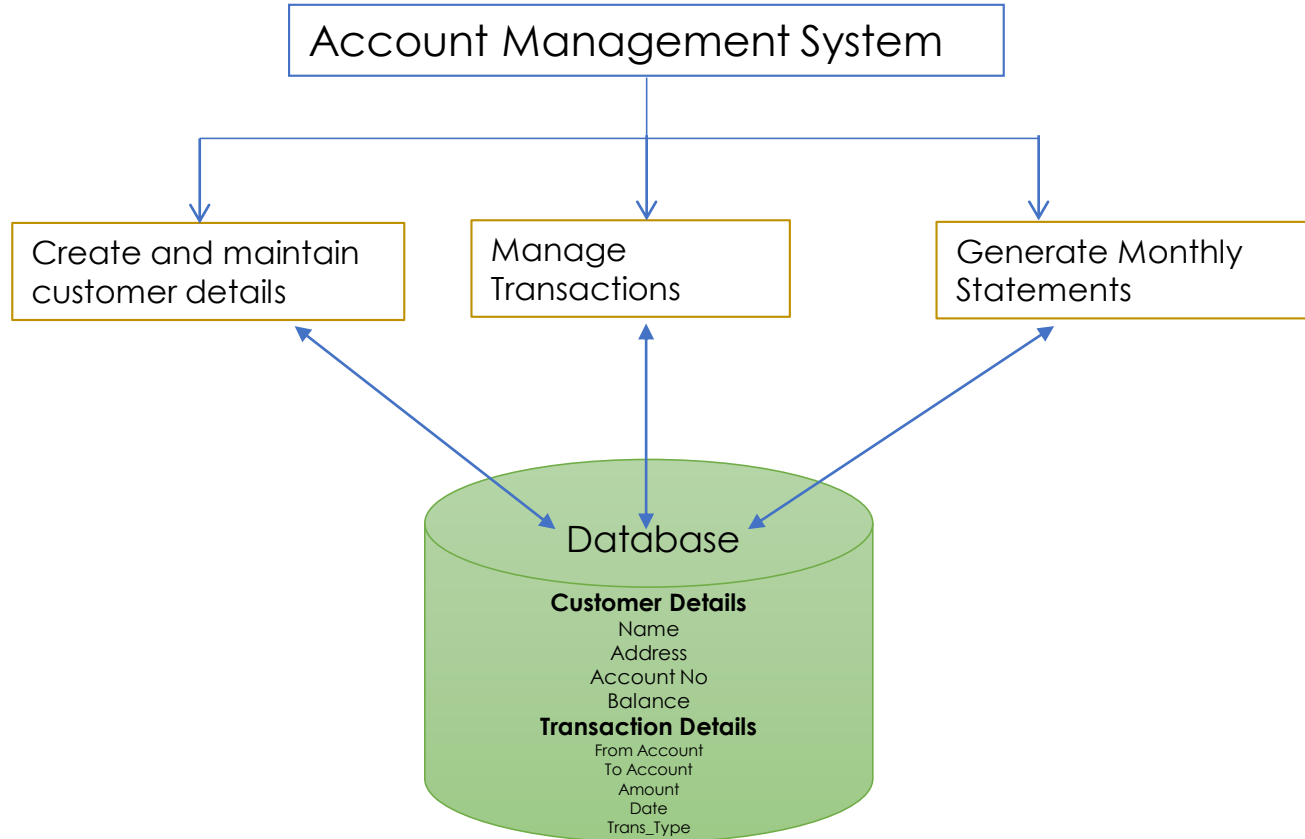
The database concept

- First design the data then write the application programs
- Manage relationships between data items independently of the application programs
- Requires
 - Database schema
 - User interface

Database

- A Database is **an integrated collection of stored operational data** that can be used by the application systems of a particular enterprise.
- Centralised storage of data, not in individual files

Data in a database system



Databases in use

- Today without databases we wouldn't be able to:

Register as a student at University	Book air travel tickets	Book Appointments with GP	Online Shopping
• Student Records • Results • Use the online course materials (VLE)	• Customer Details • Flight Details • Flight Schedules • Booking Details	• Practice Details • Doctor's Details • Appointment Details	• Product Details • Customer Details • Order Details

- Every facet of everyday life involves the use of data from databases

Objectives of a database

- To provide the enterprise/organisation with
 - **Centralised control** of its operational data
 - **Data independence**
- Ensuring
 - Corporate Data is the responsibility of a centralised function within the enterprise
 - Not the individual departments

Centralised control

Data redundancy can be reduced

Data inconsistency can be controlled

Data can & must be shared

Corporate standards can be enforced

Security restrictions can & must be applied

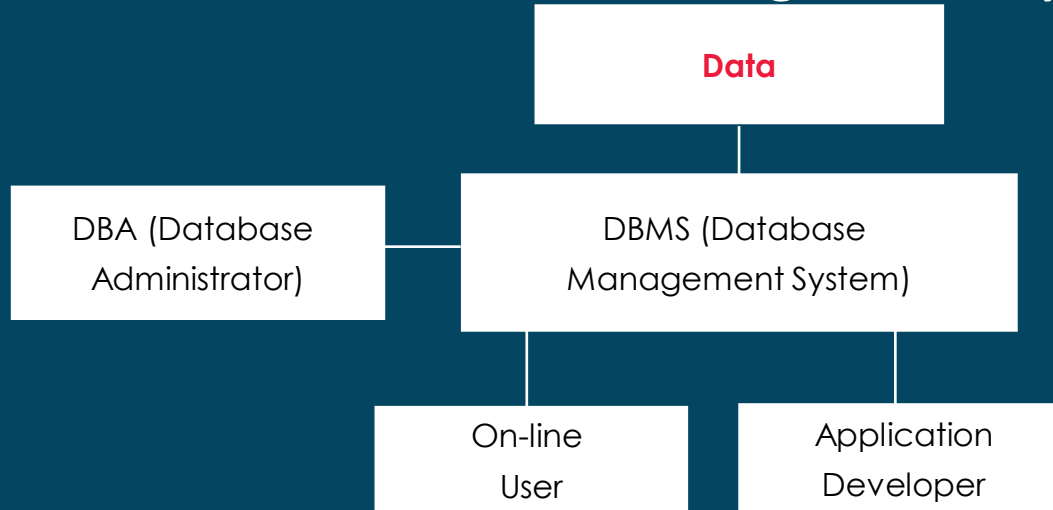
Data Integrity can be maintained

Data independence

- Database systems separate the physical data storage from the logic of the application program
- Achieved by:
 - Application program operating on the **logical data model**
 - **Database Management System (DBMS)** mapping the logical/physical transformation

What is a database system?

A database system is a repository of data administered by a database management system.



What is a DBMS?

- The DBMS consists of:



User interface



Database engine to manage the storage and access of physical data on disk



Data dictionary to record all information about the database

Summary

- Data and its role in the organization
- Data storage
 - File Systems
 - Database
- Database Management Systems
- What's Next: Relational Database Management System Overview